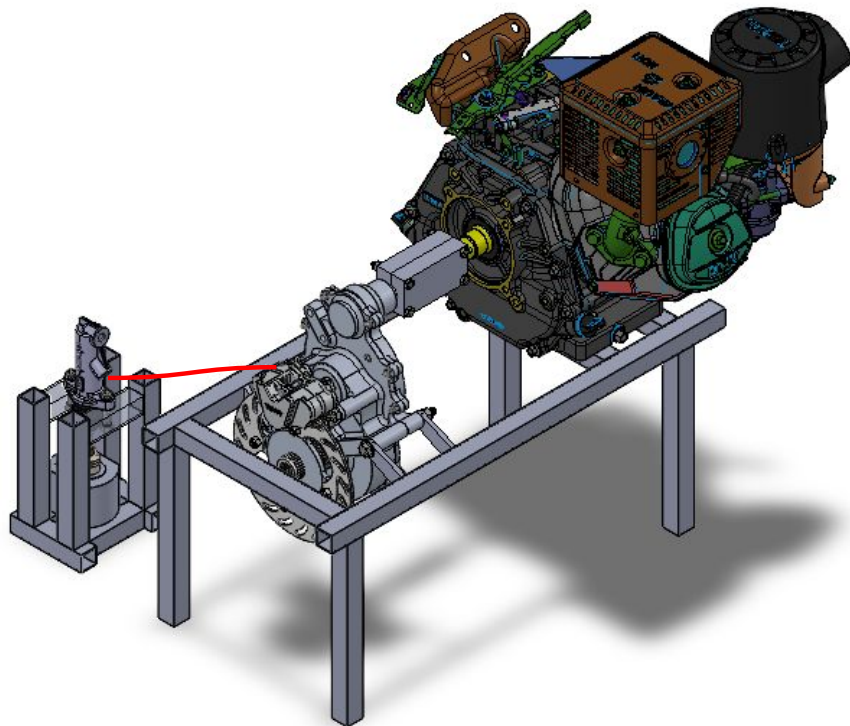


Gearbox Fatigue Life Testing

SWAYAM SHAH

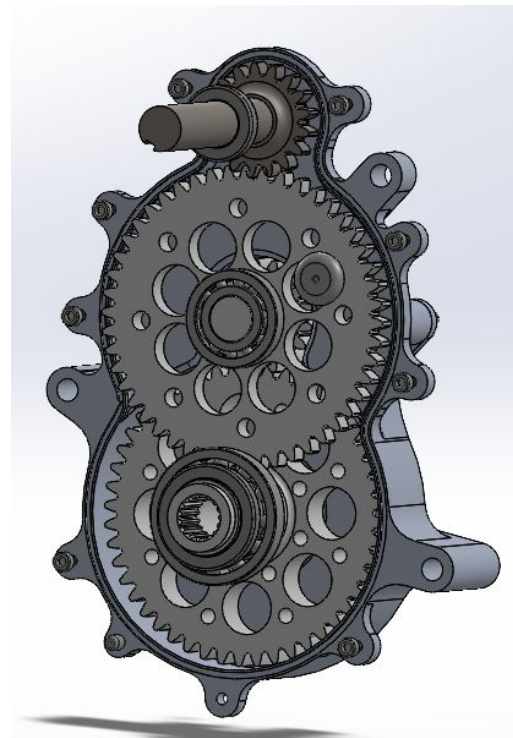
Agenda/Overview

- Executive Summary
- Test Design & Overall Stand Architecture
- Frame Structural Sizing
- Brake & Cooling System Design
- Miner's Rule Code Implementation
- Future Work



Executive Summary

- No record of gearbox fatigue failures
→ gears may be overbuilt & offer mass savings
- Gather empirical data on gear tooth fatigue failure to correlate gearbox life predictions

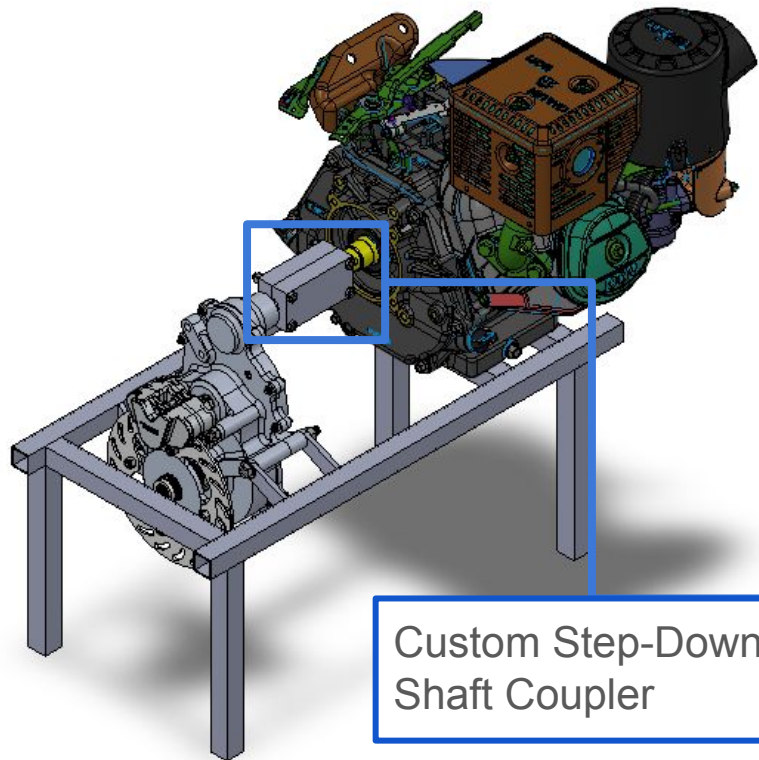


Test Design

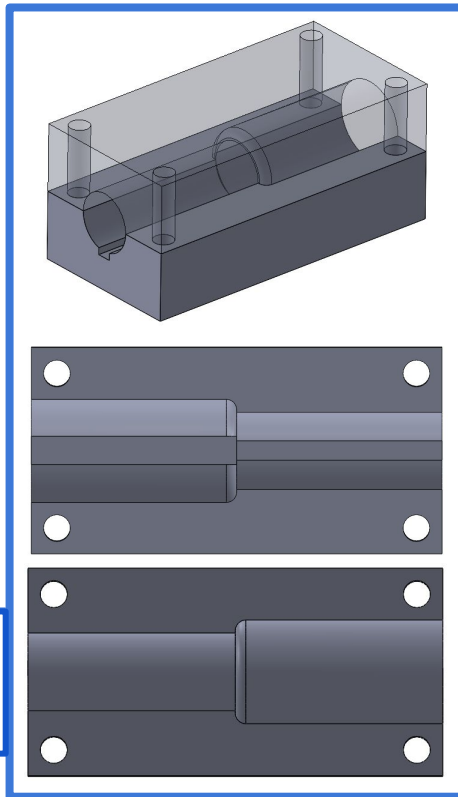
Big Picture:

- Run gearbox continuously under constant torque
 - Run engine at constant speed
 - Apply constant braking torque using disc brake
- Count cycles to failure
 - Integrate hall-effect sensor onto gearbox
- Compare accumulated damage (Miner's rule) to expected fatigue life
 - Calculate stress state in gear
 - Combine with test data (cycles to failure) to calculate equivalent damage

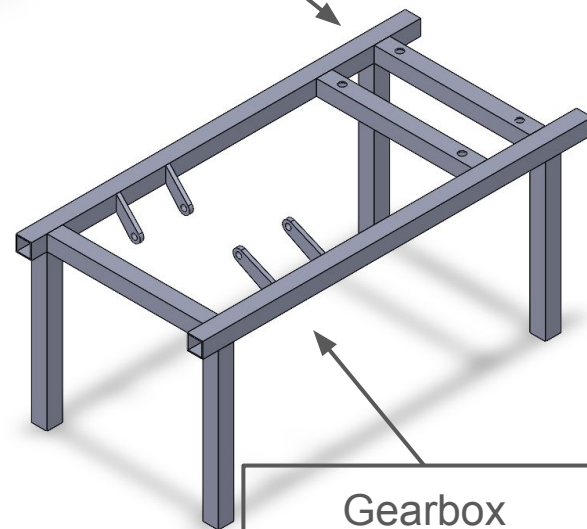
Test Stand Architecture



Custom Step-Down
Shaft Coupler

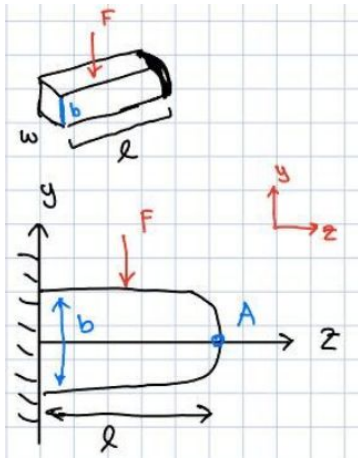
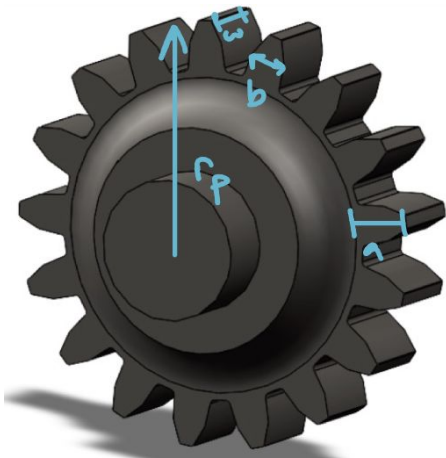


Engine Mounting Holes



Gearbox
Mounting Tabs

Input Gear Loads



Tangential Force

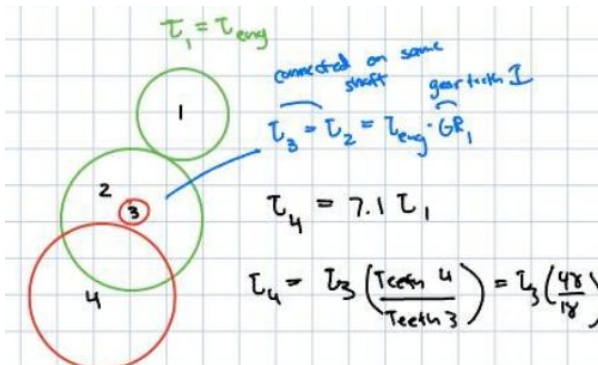
$$F_t = T(\text{engine}) / r_p$$

$$\sigma = \frac{M_y}{I} = \frac{b F l}{w b^3} = \frac{6 l F}{r_p w b^3}$$

$$T(\text{output}) = \text{Gear Ratio} * T(\text{engine})$$

$$\sigma_b = \frac{M_y}{I_{xx}} = \frac{\frac{l}{2} F}{\frac{w b^3}{12}} = \frac{l}{2} F \cdot \frac{12}{w b^3} = \frac{6 l F}{w b^3}$$

$r_p = 0.725$ in., w (thickness) = .153 in.,
 a (root) = .22 in., b (teeth thickness) = .375 in.



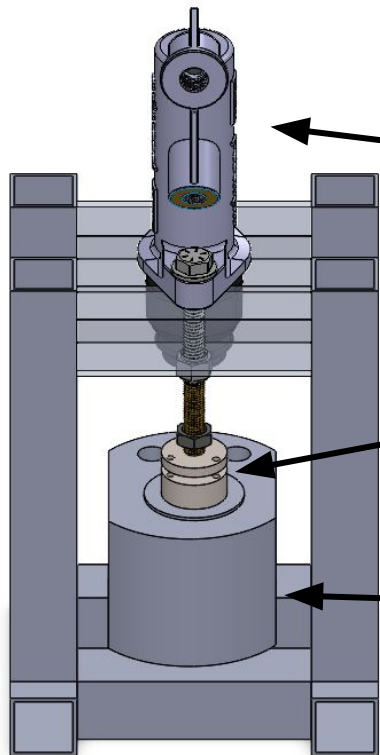
(48/18)

*

(48/18)

= 7.1:1

Brake Stand Architecture



Components:

Master Cylinder

- Pushes brake fluid through brake line to brake caliper

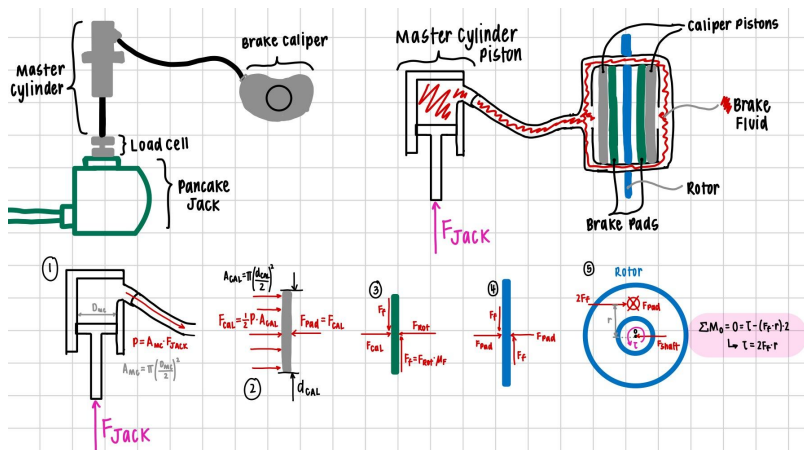
Load Cell

- Measures static force applied to master cylinder from pancake jack

Hydraulic Pancake Jack

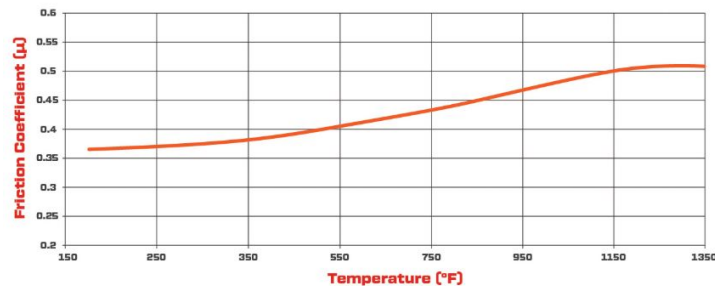
- Applies load to master cylinder

Brake System Considerations

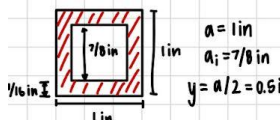
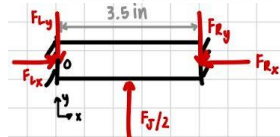


Output Torque to Pedal Weight Needed		
Quantity	Value	Unit
Output torque (gearbox shaft)	20	N*m
Rotor radius	0.0885	m
Coefficient of Friction	0.35	unitless
Rotor Friction Force	112.99435	N
Caliper Tangential Force	-112.99435	N
Rotor Force (axial)	322.841001	N
Brake Pad Force	322.841001	N
Caliper Piston Diameter	0.028519	m
Caliper Piston Area	0.00063879	m ²
Brake Line Pressure	1010788.32	N/m ²
Master Cylinder Diameter	0.01270003	m
Master Cylinder Area	0.00012668	m ²
Master Cylinder Force	128.044012	N
Weight to exert from jack	13.0523967	kg

Pedal Weight to Opposing Torque on Output Shaft		
Quantity	Value	Unit
Weight to exert from jack	13.0523967	kg
Balance Bar Force - R	128.044012	N
Master Cylinder Diameter	0.01270003	m
Master Cylinder Area	0.00012668	m ²
Brake Line Pressure	1010788.31	N/m ²
Caliper Piston Diameter	0.028519	m
Caliper Piston Area	0.00063879	m ²
Brake Pad Force	322.841001	N
Rotor Force	322.841001	N
Coefficient of Friction	0.35	unitless
Rotor Frictional Force	112.99435	N
Caliper Tangential Force	-112.99435	N
Rotor Radius	0.0885	m
Output Torque (gearbox shaft)	20	N*m

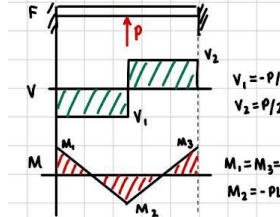


The above friction data (μ) was recorder through braking cycles from 95 mph to 40 mph at 0.5 g decel. Brakes were consecutively done until rotor temperature reached 1200°F. These graphs represent average data and are for general trend visualization only.



$$A_c = (1)^2 - (7/8)^2 = 0.234375 \text{ in}^2$$

$$Q = \frac{1^3 - (7/8)^3}{8} = 0.04216 \text{ in}^3$$



$G_y = 32,000 \text{ psi}$ (Low-carbon steel)

$F_j = 13.0524 \text{ kg} = 28.776 \text{ lb}$ (max load from jack)

$$\sigma_b = \frac{M y}{I} \text{ where } M = \frac{F \cdot L}{8} = \frac{14.388 \text{ lb} \cdot (3.5 \text{ in})}{8}$$

$M = 6.2948 \text{ lb-in}$

$$I = \frac{a^4 - a_i^4}{12} = \frac{1^4 - (7/8)^4}{12} = 0.0344848 \text{ in}^4$$

$$\sigma_b = \frac{M y}{I} = \frac{6.2948 \text{ lb-in} \cdot (0.5 \text{ in})}{0.0344848 \text{ in}^4} = 91.268 \text{ lb/in}^2$$

$$\tau_{max} = \frac{V Q}{I t} = \frac{(28.776 \text{ lb}/4) \cdot (0.04126 \text{ in}^3)}{(0.0344848 \text{ in}^4) \cdot (1/8 \text{ in})} = 68.86 \text{ lb/in}^2$$

$$\sigma_{vm} = \sqrt{\sigma_b^2 + 3\tau_{max}^2}$$

At top/bottom edges, σ_b max, zero τ

$$\sigma_{vm} = \sqrt{(91.268 \text{ lb/in}^2)^2 + 3(0)} = 91.268 \text{ lb/in}^2$$

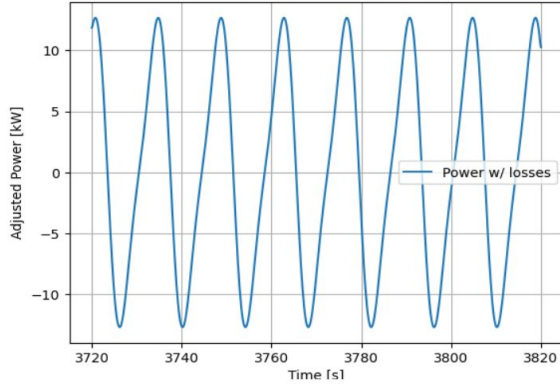
At neutral axis, zero σ_b , τ max

$$\sigma_{vm} = \sqrt{0 + 3(68.86 \text{ lb/in}^2)^2} = 119.269 \text{ lb/in}^2 = \sigma_{max}$$

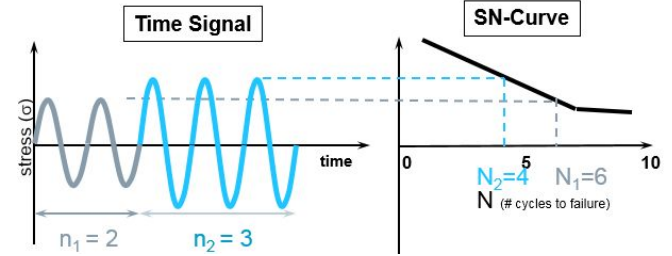
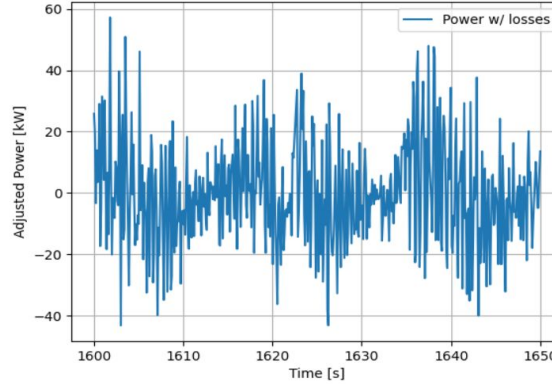
$$F.G. = \frac{\sigma_y}{\sigma_{max}} = \frac{32,000 \text{ psi}}{119.269 \text{ psi}} = 268.3$$

Code & Miner's Rule

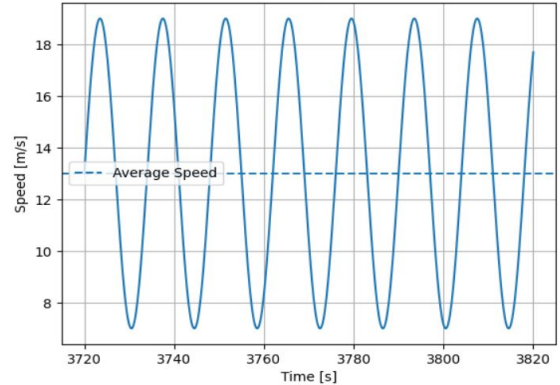
Idealized Power Profile with Losses



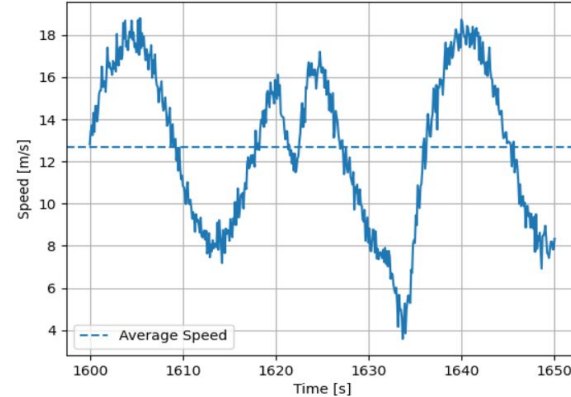
Realistic Power Profile with Losses



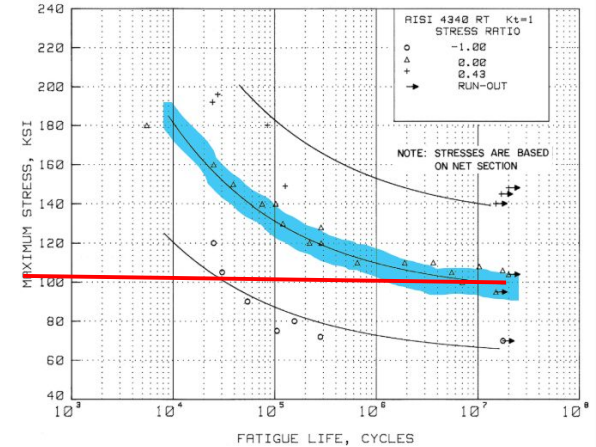
Idealized Vehicle Speed Over Time



Realistic Vehicle Speed Over Time



AISI 4340 SN Curve



Code

```
# Given stress (ksi), convert to psi (1 ksi = 1000 psi)
desired_stress_ksi = 105 # Test ksi
desired_stress_psi = desired_stress_ksi * 1000 # Convert ksi to psi

# Known gear parameters (in inches)
a = 0.22 # Distance from root to tooth (inches)
b = 0.375 / 2 # Halved face width (inches)
t = 0.153 # Tooth thickness (inches)
r_p = 0.61 # Pitch radius (inches)

# Calculate the torque needed to achieve the desired stress using the formula
torque_needed = (desired_stress_psi * b * t**3 * r_p) / (6 * a) * 7.11

# Print the calculated braking torque in Nm and in-lb
torque_needed_nm = torque_needed * 0.113 # Convert from in-lb to Nm
torque_needed_in_lb = torque_needed # in-lb is the same as calculated

print(f"Braking Torque required to achieve {desired_stress_ksi} ksi stress: {torque_needed_in_lb:.2f}")
print(f"Braking Torque required to achieve {desired_stress_ksi} ksi stress: {torque_needed_nm:.2f}")
```

Braking Torque required to achieve 105 ksi stress: 231.68 in-lb

Braking Torque required to achieve 105 ksi stress: 26.18 Nm

Code

```
# Material properties for AISI 4340 (simplified)
sigma_f_prime = 939.21 # Fatigue strength coefficient (MPa)
b = -0.09 # Fatigue strength exponent (empirical value)

# Endurance limit (MPa) – minimum stress for material to take damage
endurance_limit = 159 # MPa (based on material data)
```

```
# Total damage (simple model of applying Miner's Rule for constant torque)
damage = n_i / cycles_to_failure # Miner's Rule for constant torque (simplified)

# Determine failure prediction
if damage >= 1.0:
    print("Prediction: Gear tooth will fail!")
    print(f"Total Cycles to Failure: {cycles_to_failure:,.0f} cycles")
else:
    print("Prediction: Gear tooth still safe!")
    print(f"Remaining cycles to failure: {cycles_to_failure - n_i:,.0f} cycles")
```

```
Prediction: Gear tooth still safe!
Remaining cycles to failure: 369,136,798 cycles
```

```
=== Experimental Conclusion ===
```

```
Based on the input torque of 45 Nm (398.28 in-lb), the gear experiences a stress of 23077.92 psi (159.12 MPa).
The gear is predicted to continue operating for 369,136,798 cycles before failure.
```

Gear Tooth Sensor Implementation:

$\text{Rev/s} * (\text{hours want to run}) = \# \text{ Cycles}$

Cycles => how much T Brake to apply

Next Steps

- Build / weld test stands
- CNC machine shaft coupler
- Integrate hall effect sensor (5mm gap)
- Run FEA on high-stress components
- Run test procedure and analyze results